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Examining Potential PFAS Contamination of Private Wells from a High School in Rural Maine

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Introduction

Per- and polyfluoroalkyl substances (PFAS), or "forever chemicals," have been used for decades in consumer and industrial products to confer nonstick, moisture, and stain-resistant properties. They are increasingly found in surface water and groundwater and have been associated with impacts on human and ecosystem health worldwide.[1,2] In 2016, PFAS were discovered in the water and soil on a dairy farm in Maine. Wastewater and industrial sludge had been used as fertilizer on the farm, a practice common on Maine farms from the 1970s through the 2000s.[3] The Maine Department of Environmental Protection (DEP) tested household wells in surrounding communities and discovered many with PFAS levels exceeding the state standard of 20 ng/L total for six PFAS contaminants (hereafter referred to as ME-6): perfluorooctanoic acid (PFOA), perfluorooctane sulfonic acid (PFOS), perfluorohexane sulfonic acid (PFHxS), perfluorononanoic acid (PFNA), perfluoroheptanoic acid (PFHpA), and perfluorodecanoic acid (PFDA).

Many rural schools in Maine have detected PFAS in their drinking water, although the sources of contamination are unknown.[4] We hypothesize that, in some cases, this PFAS contamination originates from the school's septic systems (Figure 1). In 2022, PFAS was discovered in the drinking water at Mount Desert Island High School (MDIHS) in Bar Harbor, Maine. In 2023, we collected drinking water samples from neighboring homes for PFAS analysis and compared the results with those of drinking water, wastewater, groundwater, and surface water from the school campus. Here, we report those results and identify MDIHS as the likely source of PFAS contamination of domestic wells on neighboring properties.

Methods

Community Engagement

On 15 May 2023, we attended a public meeting with school officials and community members. We learned about PFAS contamination of the school's drinking water, mitigation efforts, and additional monitoring of sites on the MDIHS campus. After the meeting, we met with the school's neighbors to discuss concerns about the potential PFAS contamination of their properties. We provided PFAS drinking water sample kits. On 20 July 2023, we met with homeowners to share test results and discuss reaching additional neighbors. At a public meeting on 21 August 2023, high school officials shared the results of additional monitoring of sites on the MDIHS campus, which we used to compare with homeowner drinking water PFAS test results.

Sample Collection, Quality Control, and Analyses

Drinking water samples were collected from eight houses (numbers 1–6, 8, and 9) by the homeowner or us. We also collected two wetland samples near the school (Figure 2A). Water sample kits were provided by Maine Laboratories, which conducted analyses using a modified US Environmental Protection Agency (EPA) 537 method[5] with isotope dilution to assess the concentrations of 28 PFAS. Each sample kit contained four labeled bottles in a PFAS-free Ziploc bag, three empty bottles, and one with PFAS-free water. The sampler was instructed to wash and dry hands and wear the PFAS-free gloves provided. They filled one bottle with drinking water after running the faucet for 5 min, took a duplicate sample immediately afterward, and poured a PFAS-free solution into the third empty bottle to check for contamination during sampling. All samples were returned to the plastic bag, refrigerated, and shipped on ice to Maine Laboratories, either by the homeowner directly or after being transported by the homeowner on ice to our lab for shipping, with a chain of custody sheet signed by the sampler and shipper. A duplicate sample was analyzed when the original samples were ≥20 ng/L for ME-6. There were no detection events in the field blanks. Quality controls for every batch were a lab reagent blank and a lab control sample. Method detection limits (MDLs) ranged from 0.46 to 1.18 ng/L, depending on the analyte.

Consulting firm Haley Ward collected groundwater, surface water, and wastewater samples on the MDIHS campus between 18 April and 7 June 2023. Alpha Analytical analyzed those samples using a modified EPA 537 method[5] with isotope dilution. Field blanks and duplicate samples were not collected. MDIHS custodial staff collected the drinking water sample on 2 May 2022, which was analyzed by Alpha Analytical using EPA 533.[6] The owner of house #7 collected their drinking water sample on 8 August 2022, which was also analyzed at Alpha Analytical. Alpha Analytical used a method blank with a spiked internal standard, a lab control sample, a matrix spike, and a lab duplicate for quality control but did not report method detection limits.

Data Reporting, Sharing, and Analysis

Maine Laboratories reported results to individual homeowners via email. With agreement from project participants, these data were shared through an interactive map (created using ArcGIS Pro 3.2.0) and on the citizen and community science data portal, Anecdata, developed at the MDI Biological Laboratory.[7] We

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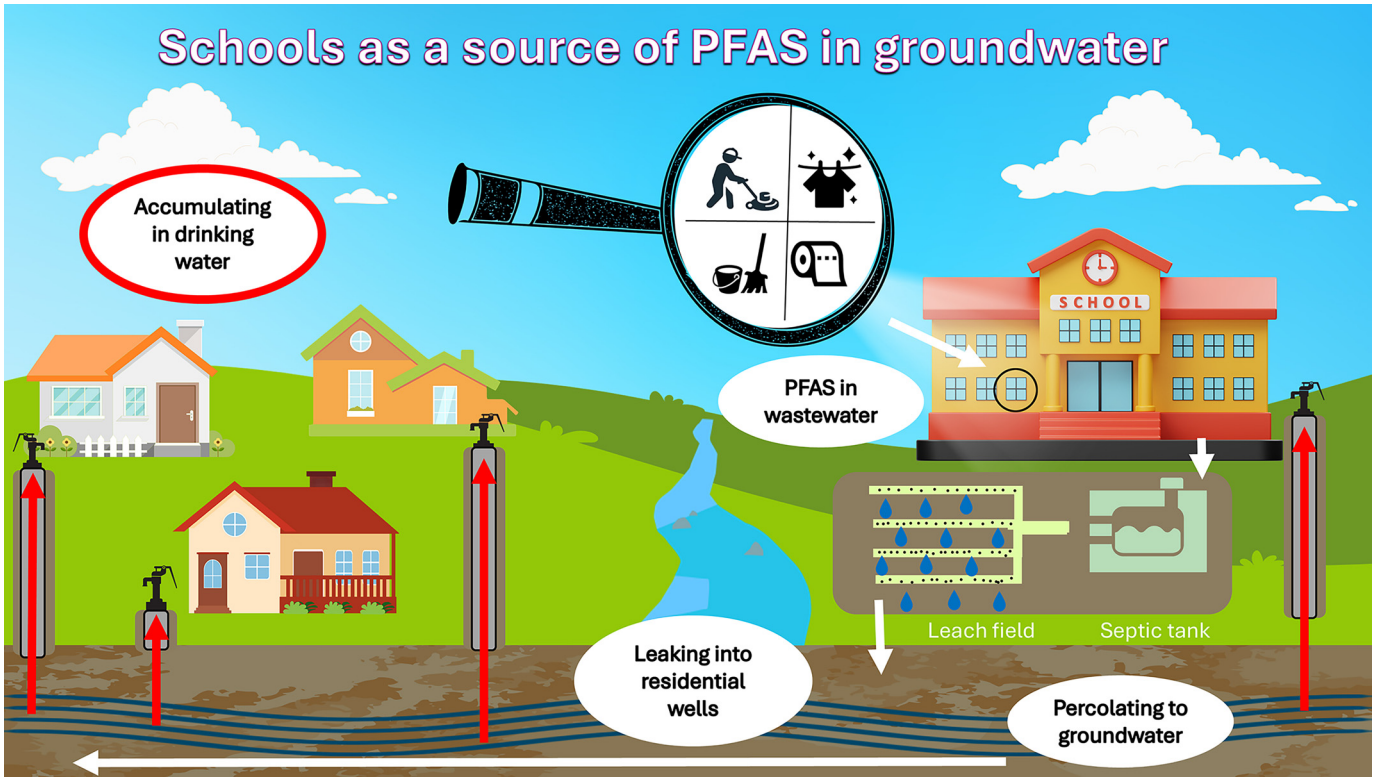


Figure 1. Rural schools often have septic systems and leach fields for handling waste. Wastewater containing PFAS-laden compounds can travel from the septic tank to the leach field and percolate into groundwater, which, in some rural areas, is the source of drinking water for the school and nearby neighbors. We hypothesize that some schools have "self-contaminated" the groundwater, leading to PFAS in their drinking water and wells on neighboring properties. Mount Desert Island High School (MDIHS) (which has three septic ponds serving a similar function as a leach field) and surrounding properties provide an example of where this has likely occurred in rural Maine. Note: PFAS, per- and polyfluoroalkyl substances.

conducted statistical analyses and graphing using R version 4.3.2. Spearman's rho correlation coefficients were calculated between pairs of sites based on the concentrations of nine PFAS (only PFAS found in at least three samples were included in the analyses). Nondetects were treated as zeros. The original data are archived in the Environmental Data Initiative Data Portal (Identifier edi.1630.5).[8]

Results and Discussion

Our comparison of water samples from households ($n=8$) and wetlands ($n=2$) surrounding MDIHS to drinking water, groundwater, surface water, and wastewater from the school campus reveals similar PFAS profiles between the samples. Figure 2A depicts the sites where samples were collected: Only houses #2 and #3 exceeded the 20 ng/L ME-6 standard, at 24.4 and 64.1 ng/L, respectively.

PFAS levels and correlations generally decreased as distance increased from the MDIHS campus, except for house #1, which may not share the same groundwater, and house #7, which is on the receiving end of surface and groundwater flow from the high school property. PFAS profiles from the school and wetland samples were highly correlated (average $r=0.68\pm0.11$) with drinking water samples from each surrounding property except house #5. Comparisons of concentration and fractional profiles of household PFAS with those from MDIHS (Figures 2C,D) suggest that the contamination in the household wells came from MDIHS. With the discovery of multiple private wells with elevated PFAS levels near MDIHS, the Maine DEP has investigated and assisted homeowners with mitigation strategies, including installing whole-house filtration systems where necessary.

Because PFAS detected in our water samples included seven of eight chemicals identified in commercial floor wax products,[9] floor wax application in the school may have contributed to PFAS contamination in surrounding neighborhoods. However, other school items, such as toilet paper, cleaners, stain-resistant upholstery, and carpets, also contain PFAS and could have contributed to contamination.

Data from the Maine Drinking Water Program reveal that of 196 schools tested as of November 2024, 36 (18.4%) exceeded ME-6 and sixty-one (31%) exceeded new standards released by EPA in April 2024, on at least one occasion. Other communities may benefit from our community-engaged research approach, which supports residents in determining PFAS contamination of their drinking water while facilitating local public health interventions by state agencies. Given the scope of this problem and its implications for public health, a broader statewide strategy may be more prudent. Predictive modeling based on contamination source, soil, hydrology, groundwater, and other variables should be used to identify households likely to have elevated PFAS levels to facilitate early testing, intervention, and mitigation strategies.[10]

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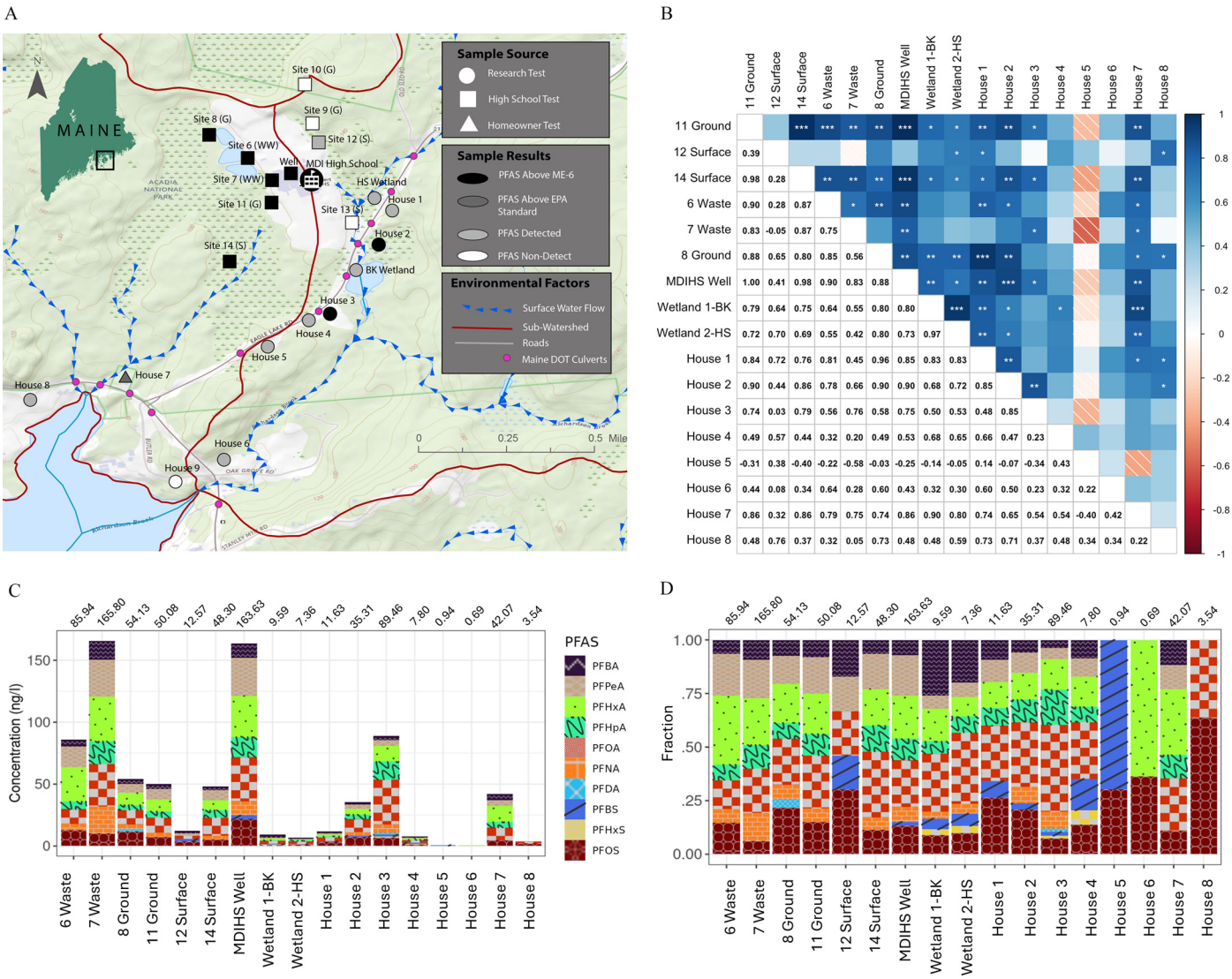


Figure 2. (A) The map depicts the locations on the Mount Desert Island High School (MDIHS) campus, nearby wetlands, and houses where water was sampled to analyze per- and polyfluoroalkyl substances (PFAS). The campus site numbers correspond to numbers in the dataset supplied by MDIHS. Two of the water samples analyzed from these sites were school wastewater samples (WW), both of which had PFAS above 20 ng/L for perfluorooctanoic acid (PFOA), perfluorooctane sulfonic acid (PFOS), perfluorohexane sulfonic acid (PFHxS), perfluorononanoic acid (PFNA), perfluoroheptanoic acid (PFHpA), and perfluorodecanoic acid (PFDA), referred to in this paper as ME-6. Site 7 is the influent (wastewater) coming directly from the school. Site 6 is a pump house located at one of the septic ponds used to spray playing fields on the school property during the summer months. Four of the water samples analyzed were groundwater samples (G), two of which had detectable PFAS. Three were surface water samples (S), two of which had detectable PFAS. The sites designated by round symbols, identified as "Research Test" in the legend, were sampled by homeowners or researchers as part of this study. The sites designated by square symbols, identified as "High School Test" in the legend, were sampled by the high school's environmental engineering consulting firm, Haley Ward, or the custodial staff at MDIHS. House 7, designated by a triangle, was sampled by the homeowner before this study and is identified in the legend as "Homeowner Test." (B) The heat map includes PFAS test results from MDIHS water samples designated as waste, ground, surface, and well, research test sites designated as wetlands (BK, brook; HS, High School), and houses. Samples from one house and three high school campus test sites had no detectable PFAS; only those water samples with detectable PFAS were included in the analysis. Spearman's rank correlation (rho) was used to compare PFAS concentration rankings between samples. The heat map displays the coefficients for correlations on the left side; the *p*-value is symbolized by asterisks on the right side, defining significance as follows: *, <0.05; **, <0.01; ***, <0.001. The darker the color, the higher the correlation. The bluer the color, the more positive the correlation. The diagonal lines on the orange cells of the heat map facilitate the distinction between the two colors. Only analytes that were detected in at least three sampled sites were used to generate the heat map: PFOA, PFOS, PFHxS, PFHpA, PFNA, perfluorobutanoic acid (PFBA), perfluoro-n-pentanoic acid (PFPeA), perfluorohexanoic acid (PFHxA), and perfluorobutane sulfonate (PFBS). (C,D) Concentrations (C) and fractional profiles (D) of different types of PFAS for MDIHS water sample sites, nearby wetlands, and household wells near MDIHS. The graphs include PFAS test results from MDIHS water samples designated as waste, ground, surface, well, research test sites designated as wetlands (BK, brook; HS, High School), and houses. The numbers above the bars on both plots indicate the total concentration of the 10 PFAS depicted. Only those samples with detectable PFAS were included in these graphs.

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